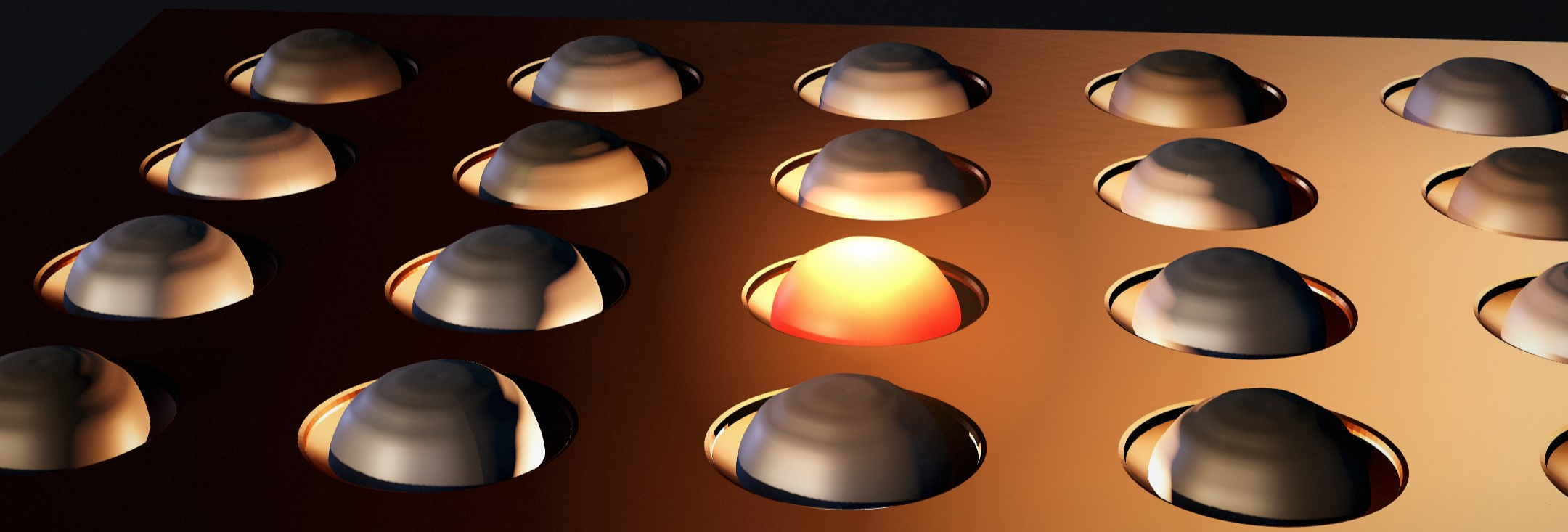
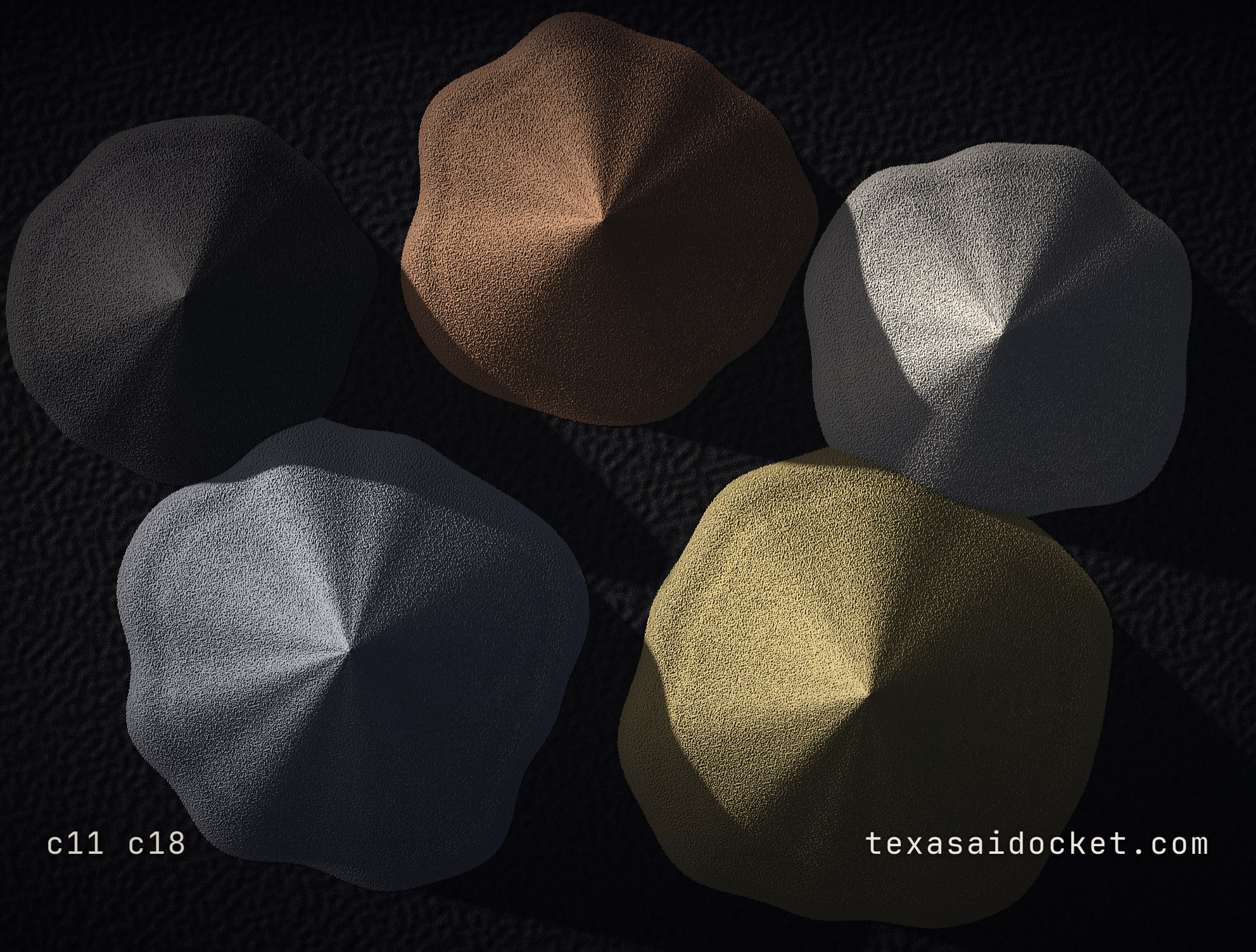
The metal still has to be melted

Texas A&M plans the Aggie AI Foundry, where computer controlled arms will mix, melt and test new alloys. A $2 million WoodNext Foundation grant will help build it. Drawn to illustrate.



Roughly a trillion ways to mix them

Several dozen elements give AI roughly a trillion possible mixes, the university says. Existing self driving labs focus on materials that are relatively easy to make.
Drawn to illustrate.



c11 c18

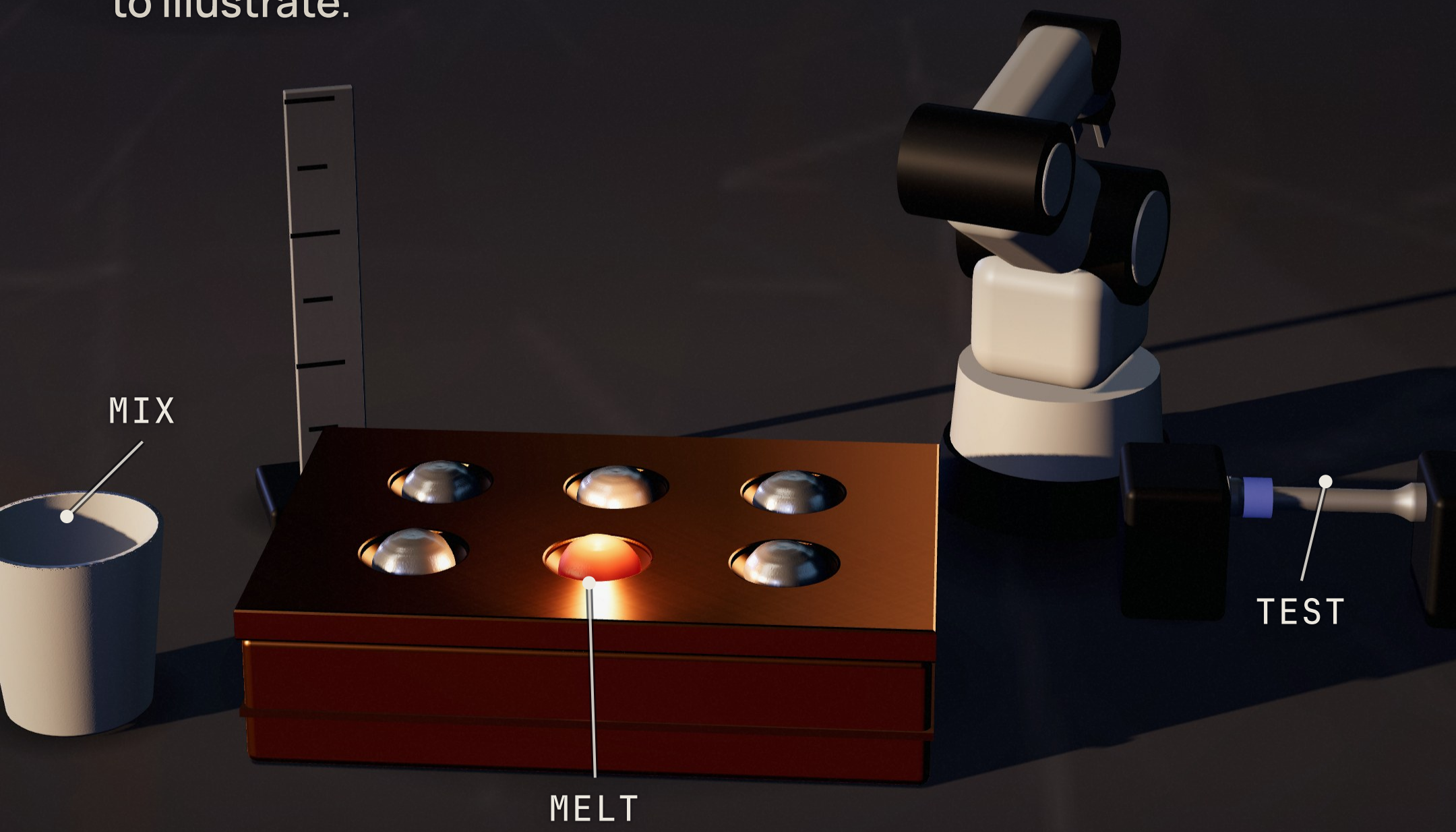
$2 million, in buttons

The WoodNext Foundation manages the philanthropy of Roku's founder, Anthony Wood, and his wife Susan. The grant comes through a donor advised fund. One button is $100,000, so 20 buttons. Drawn to illustrate.



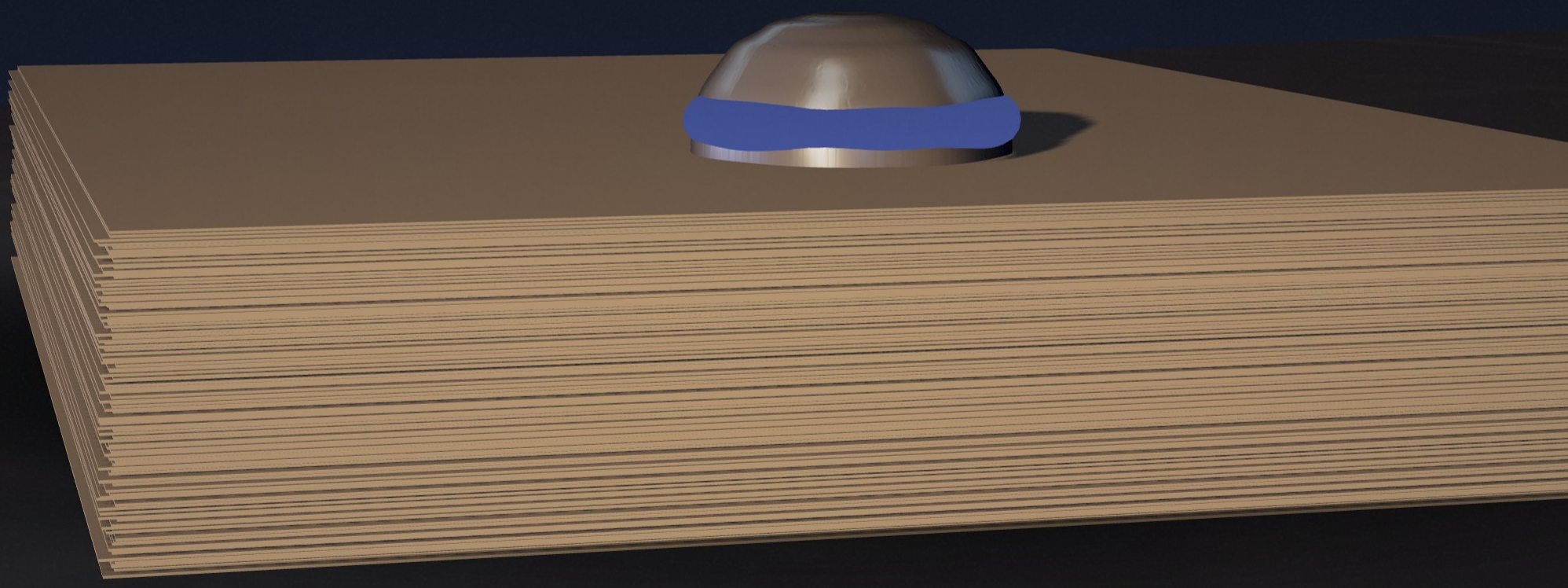
Mix, melt and test, on one line

Robotics will mix, melt and test the materials with computer controlled arms along an assembly line that takes samples, the university says. It says existing self driving labs can't handle alloys at up to 3,000 degrees Celsius. Drawn to illustrate.



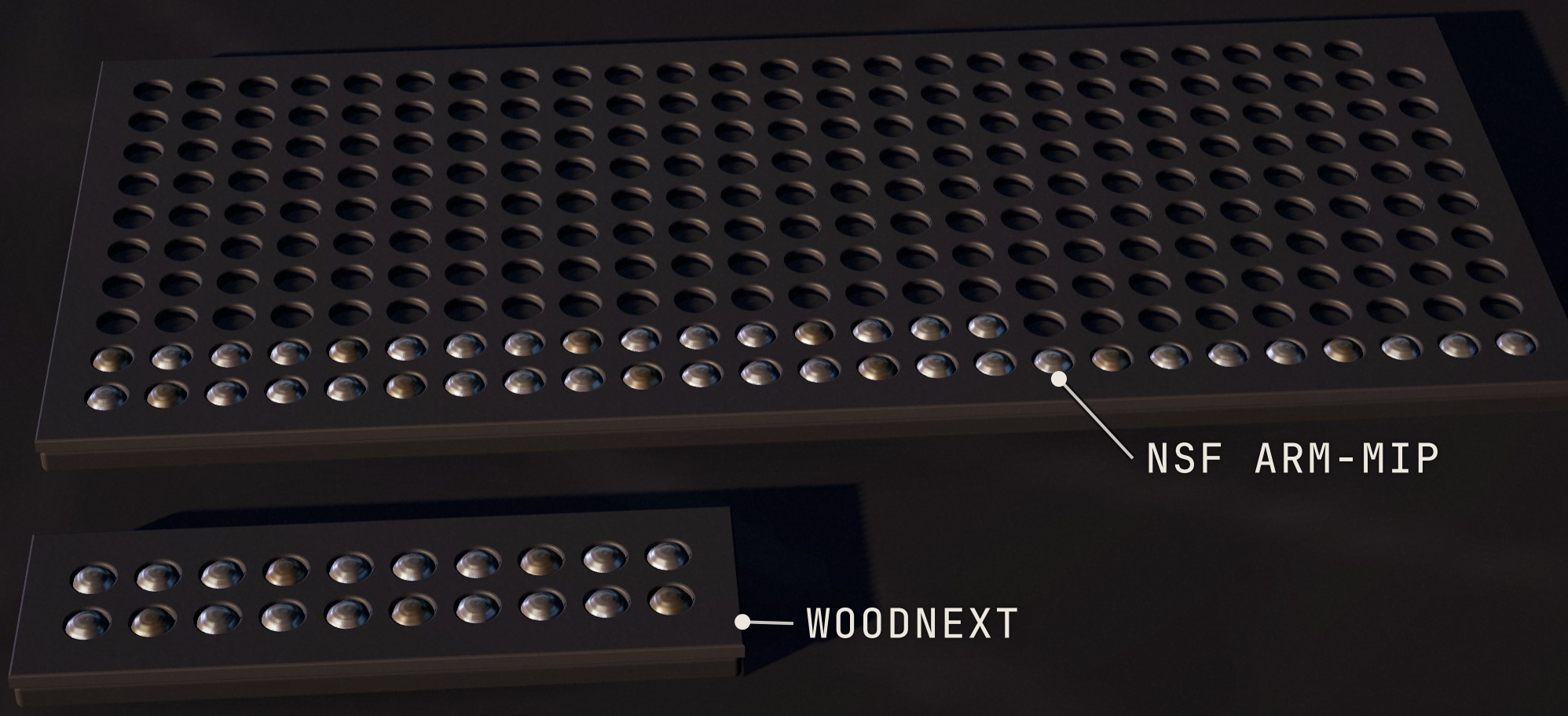
Two researchers, more than 100 papers

Ibrahim Karaman, who heads the department, and Raymundo Arróyave conceived the foundry. They have published more than 100 papers together. The university calls it the first lab of its kind in academia.
Drawn to illustrate.



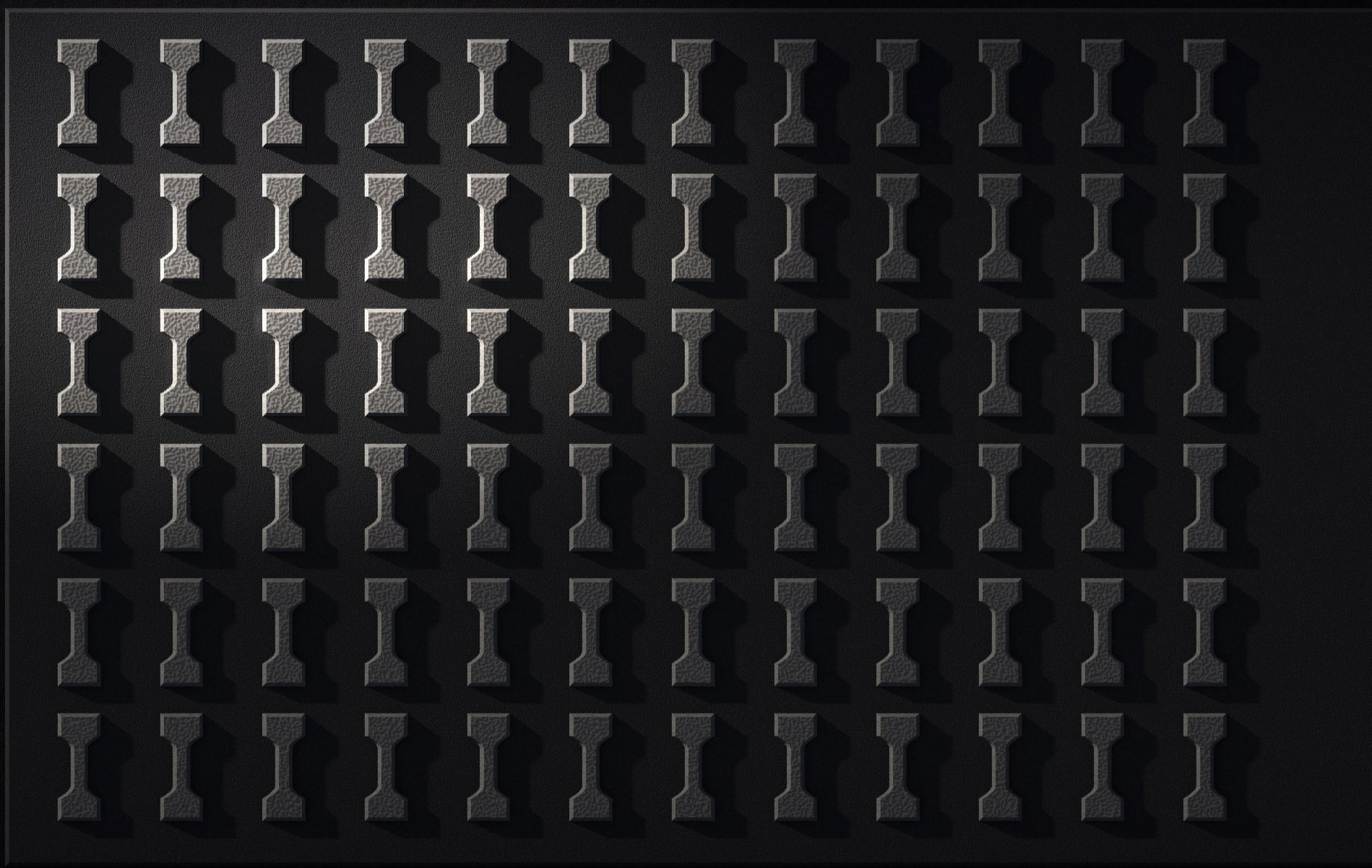
NSF has obligated $4.1 million of $24.9 million

The $24.9 million is the award's estimated total. Its Autonomous Robotic Metallurgist at Texas A&M is built to design, make, test and learn from new alloys. In front, the WoodNext grant at $100,000 a button. Drawn to illustrate.



Six years, one bar a month

The NSF award runs from September 1st to August 31st, 2032. The platform is open access, and its intended users include researchers at non-R1 and EPSCoR institutions. One bar is one month. Drawn to illustrate.



c21 c23 c27 c28

No computational tool can predict it, Arróyave says

Arróyave says no computational tool anywhere in the world can predict how an alloy is going to behave. His NSF backed PhaseForge applies machine learning to CALPHAD to say whether a material can exist at all. Drawn to illustrate.



c17 c29 c30 c32 c33

So the metal still gets tested

Texas A&M's goal is alloys in a week or less instead of months, a target and not a result. PhaseForge is free online and anyone can contribute to it. The NSF platform is open access. Drawn to illustrate.

